

FastAPI Developer Technical Assessment

This assessment consists of two independent FastAPI applications. The goal is to evaluate application design, authentication, authorization, API development, scalability, performance, and code quality.

Application 1 – Identity Service

User registration User login JWT access token generation Refresh token support User profile endpoint Role-based access control (Admin/User) PostgreSQL database integration API documentation using OpenAPI/Swagger

Application 2 – Client Application

Uses Application 1 as the authentication provider Users authenticate through Application 1 Protected endpoints must validate issued tokens User information retrieval from the Identity Service Proper error handling for invalid or expired tokens

Additional Functional Requirements

Task A – Analytics Service

Create an endpoint that accepts a numerical range and returns: Total count of matching results
Execution duration Structured JSON response The implementation should be able to handle large input ranges efficiently.

Task B – Data Aggregation Service

Create an endpoint that retrieves data from multiple external services and returns a combined response. Requirements: Fetch data from at least three external APIs Handle failures gracefully Provide meaningful error messages Return a unified response structure

General Requirements

Python 3.11+ FastAPI PostgreSQL SQLAlchemy or equivalent ORM Environment-based configuration Docker support Unit tests Logging and error handling Clean project structure README with setup instructions

Evaluation Criteria

Code quality and maintainability Architecture and design decisions Security implementation
Performance considerations Testing coverage Documentation quality Error handling and resilience